

AEC INFORMATICS

ITI – Architecture, Engineering, and Construction Informatics (AECI) Specialization - is a product-based program that will empower you to learn how to utilize state-of-the-art information and communication technology tools to develop solutions that tackle real-world problems as they arise in the Architecture, Engineering and Construction (AEC) industry.

JOB PROFILES

AEC Informatics Specialist • AEC Software Developer • BIM Automation Developer



TOTAL HOURS

1,308hrs

Theory + hands-on labs



DELIVERY

Blended

Online & on-site



UNIVERSITY GRADE

Good

Minimum grade required



CERTIFICATIONS

2

Industry-certified



Where graduates work



Certifications earned

-  **CAD/BIM Field** — Variety of available Autodesk Certified Professional certificates
-  **SWD Field** — Variety of certified professional certificates for the studied tools



Selection process

- 01 IQ & English Exam**
 Online exam covering problem-solving aptitude and English proficiency.
- 02 Technical Exam**
 Computer-based assessment covering the program's core technical fundamentals.
- 03 Technical Interview**
 Interview with the technical committee.
- 04 Interpersonal Interview**
 Candidates who pass the technical interview are invited to this final stage.



Who may apply

All University Graduates with prerequisites of bachelor's degree in engineering discipline related to AEC Industry

- o Architectural Engineers
- o Civil/Structural Engineers
- o Mechanical Engineers
- o Electrical Engineers
- o Plumbing Engineers

Program content — 1,308 hours

CATEGORY	HOURS	SHARE	KEY COURSES
● Fundamental Courses	404 hrs	31%	Operating Systems Fundamentals, Computer Networks Fundamentals, Database I - Database Fundamentals, Database II - Advanced Topics in Database, Programming I - Programming Fundamentals (with C#), Programming II - Object-Oriented Programming (with C#), Programming III - Advanced Topics in Programming (with C#), Data Structures and Algorithms (with C#)
● Core Courses	504 hrs	39%	Desktop Application Development (with C# & .NET), Web Development I – Front-End Development, Web Development II – Back-End Development, Software Architecture and Design, Software Testing Fundamentals, Scientific Programming with C/C++, Scientific Programming with Python, Scientific Computing II - Numerical Tools & Algorithms, Selected Topics in Engineering Informatics, Computer Graphics I - Mathematics Foundation, Computer Graphics II - Programming with OpenGL, Computer Graphics III - Programming with WebGL, Building Information Modeling I - Fundamentals, Building Information Modeling II - Design Studio, Building Information Modeling III - Development Studio, Geographic Information Systems I - Fundamentals, Geographic Information Systems II - Geodatabases, Geographic Information Systems III - Development, Artificial Intelligence I - AI Fundamentals, Artificial Intelligence II - Generative AI
● Soft Skills	100 hrs	8%	Communication Essentials for Professionals, High Impact Presentations, Job Seeking Skills, Professional Demeanor, Progressive Teamwork (Workshop)
● Graduation Project	300 hrs	23%	

Hands-on labs are conducted in-person. Online components cover theory and self-study modules on MaharaTech.

Student Deliverable

Each student must deliver at least ONE freelancing job and optionally, a professional certificate based on the student's engineering discipline track.

Before you apply

Explore these resources before your interview. Applicants are expected to have reviewed this material.

- Knowledge of computer programming and software development.
- Academic/Professional experience in the field of CAD/BIM for AEC Industry.
- ITI Values & Mission
<https://iti.gov.eg/about-us>
- More References
<https://www.youtube.com/@AECInformatics>

APPLY AT

<https://iti.gov.eg>

Applications open each intake period. Check the website for current dates and deadlines.